M5 — Mel front-end parity and the neural tangent kernel

jax-experiments writeup · M5

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Status: done (N7 and O7) ·
Hardware: CPU, 4 cores, x86_64. No GPU was used ·
Data: `data/n7-kernel.json`, `data/n7-train.json`, `data/n7-alpha.json`, `data/n7-alpha-gelu.json`, `data/n7-converged.json`, `data/n7-floor.json`; O7: `data/o7-stages.json`, `data/o7-decode.json`, `data/o7-controls.json` ·
Reproduce: `uv run python experiments/n7_ntk/run.py kernel train alpha alpha-gelu converged floor plot` (about 25 minutes; needs `../ml-ladder` for B1's moons generator; MNIST downloads once into `data/`). O7: `cargo build --release` in `tools/candle_mel`, then `experiments/o7_mel/run.py stages decode controls plot` in its own environment (command in the script's docstring; about 5 minutes; needs `../ml-ladder` for the recorded transcripts)

N7 — The empirical neural tangent kernel

Question

How well does the infinite-width neural tangent kernel (NTK) predict what a finite network actually does, as width grows? And when does feature learning break the agreement?

The idea

Training is linear in function space. A network $f(x; w)$ trained by gradient descent on squared loss moves its outputs as

$$\dot{f} = -\eta\,\Theta_w(f - y)/n, \qquad \Theta_w(x, x') = \langle \nabla_w f(x), \nabla_w f(x') \rangle.$$

At infinite width the kernel freezes (Jacot et al., 2018). In the NTK parameterisation, Θ_w converges to a deterministic kernel Θ_∞ at initialisation, and stays there during training. The network then *is* kernel regression with Θ_∞, a prediction made without training a network at all.

Finite width breaks this two ways.

- At initialisation, Θ_{w_0} fluctuates around Θ_∞, with a relative error of order $1/\sqrt{\text{width}}$.
- During training, Θ_w moves. That movement is feature learning, and the kernel description fails exactly when it is large.

A knob separates the two (Chizat, Oyallon & Bach, 2019). Train the centred, scaled model $g = \alpha\,(f(w) - f(w_0))$ on $\frac{1}{2\alpha^2}$ times the squared loss:

- Its function-space dynamics at $t = 0$ are the same for every α.
- To move g by $O(1)$, the weights move by $O(1/\alpha)$.
- Large α is therefore "lazy": the kernel should stay put, and g should track its own linearisation with an error of $O(1/\alpha)$. Small α forces the weights, and the features, to move.

The math

Network. Two hidden ReLU layers and a scalar output, NTK parameterisation, $\sigma_w^2 = 2$, $\sigma_b^2 = 0.1$:

$$h_1 = \sigma_w W_1 x/\sqrt{d} + \sigma_b b_1, \quad h_l = \sigma_w W_l\,\phi(h_{l-1})/\sqrt{n} + \sigma_b b_l, \quad f = \sigma_w w^\top \phi(h_L)/\sqrt{n} + \sigma_b b.$$

Analytic NTK (Jacot et al.; the arc-cosine kernels of Cho & Saul, 2009). For (u, v) Gaussian with covariance $\begin{pmatrix} k_{11} & k_{12} \\ k_{12} & k_{22} \end{pmatrix}$ and $\theta = \arccos\left(k_{12}/\sqrt{k_{11}k_{22}}\right)$:

$$\mathbb{E}[\phi(u)\phi(v)] = \tfrac{\sqrt{k_{11}k_{22}}}{2\pi}\big(\sin\theta + (\pi - \theta)\cos\theta\big), \qquad \mathbb{E}[\phi'(u)\phi'(v)] = \tfrac{\pi-\theta}{2\pi}.$$

The recursion runs through the layers:

$$\Sigma^l = \sigma_w^2\,\mathbb{E}[\phi\phi](\Sigma^{l-1}) + \sigma_b^2, \qquad \Theta^l = \Theta^{l-1}\,\sigma_w^2\,\mathbb{E}[\phi'\phi'](\Sigma^{l-1}) + \Sigma^l.$$

Empirical NTK, without the Jacobian. For a dense layer, $\partial f/\partial W_l = \delta_l\, a_{l-1}^\top$, an outer product, where $\delta_l = \partial f/\partial h_l$ comes from one vector-Jacobian product. So

$$\Theta_w = \sum_l (\Delta_l \Delta_l^\top) \odot (\sigma_w^2 A_{l-1} A_{l-1}^\top/\text{fan}_l + \sigma_b^2) + \sigma_w^2 A_L A_L^\top/n + \sigma_b^2,$$

at $O(N^2 \cdot \text{width})$ cost instead of $O(N^2 \cdot \text{params})$.

Kernel prediction after t gradient-descent steps. With step η and $g_0 = 0$, the prediction is exact for discrete steps:

$$g_t(X_*) = \Theta_{*X}\,\Theta_{XX}^{-1}(I - (I - \eta\Theta_{XX}/n)^t)\,y.$$

Two kernels go into it:

- the network's **own** Θ_{w_0}, giving the *linearised model*;
- Θ_∞, giving the *infinite-width prediction*.

What we measure and why

Tasks. Targets are ± 1 and the output is scalar, so every kernel is exact. Seeds 0–2 vary both the data draw and the initialisation.

- **Two moons**: ml-ladder B1's generator, noise 0.18, 256 training and 1,024 test points, standardised.
- **MNIST, even vs odd**: B2's normalisation, 1,000 training and 1,000 test images, 100 per digit.

Measurements.

- **Kernel at initialisation.** $\|\Theta_{w_0} - \Theta_\infty\|_F/\|\Theta_\infty\|_F$ on 256 inputs, for widths 32 to 8,192. Theory predicts a slope of $-\frac{1}{2}$.
- **Width sweep** ($\alpha = 1$; widths 64, 256, 1,024, 2,048). Full-batch gradient descent with $\eta = n/\lambda_{\max}(\Theta_\infty)$: 1,000 steps on moons, 500 on MNIST. At log-spaced steps, compare the test predictions with the linearised model and with the infinite-width prediction. Also record the **kernel drift** $\|\Theta_{w_T} - \Theta_{w_0}\|/\|\Theta_{w_0}\|$ on the training set.
- α **sweep** (width 512, α from 0.03 to 100), with the same quantities. The same η serves every α, because the initial dynamics do not depend on it.
- **A smooth control.** The same sweep with exact-erf GELU in place of ReLU. The linearised comparison needs only the empirical kernel, which autodiff gives for any activation.
- **Two controls added after reading the first results.**

  - The infinite-width kernel model trained to convergence, to tell "better features" from "further along".
  - Float32 against float64 at large α, to tell a precision floor from a real effect.

What I built

- **`src/jaxexp/ntk.py`**:
  - the analytic NTK and NNGP kernels;
  - the empirical NTK by the per-layer identity, for any activation;
  - the brute-force $JJ^\top$, used only to check it;
  - the exact discrete-time kernel-regression path.
- **`src/jaxexp/data.py`**: MNIST from Google's CVDF mirror, each file checked against a recorded SHA-256. Their MD5s are LeCun's originals.
- **`experiments/n7_ntk/run.py`**: the six parts and the figure, checkpointing every run in `runs/`.
- **`tests/test_n7_ntk.py`**, 4 tests:
  - the per-layer NTK equals $JJ^\top$;
  - the empirical NTK approaches the analytic one with width;
  - the analytic diagonal equals its closed form;
  - the kernel path starts at 0, takes one exact step, and interpolates in the limit.

Checks against independent code.

- The per-layer empirical NTK equals brute-force $JJ^\top$ to 2.8×10^{-7}.
- The analytic kernels equal `neural-tangents` 0.6.5's, to 1.2×10^{-8} (NTK) and 7×10^{-15} (NNGP), on the same 64 MNIST images in float64. `neural-tangents` no longer imports under JAX 0.11, so it ran in a separate environment with JAX 0.4.30.

Results

The kernel at initialisation (relative Frobenius distance from Θ_∞, median of seeds):

width	32	256	2,048	8,192	slope, widths 256–8,192
moons	0.378	0.088	0.039	0.024	−0.41
MNIST	0.453	0.134	0.053	0.028	**−0.48**

Width sweep at $\alpha = 1$, measured at the end of training (median and range):

task	width	net vs linearised	net vs infinite-width	kernel drift	test accuracy	∞-width NTK, same step
moons	64	0.123 (0.112–0.164)	0.085	0.301 (0.217–0.521)	0.970	0.970
moons	2,048	0.047 (0.045–0.049)	0.038	0.019 (0.018–0.020)	0.970	0.970

task	width	net vs linearised	net vs infinite-width	kernel drift	test accuracy	∞-width NTK, same step
MNIST	64	0.201 (0.201–0.211)	0.314	0.192 (0.165–0.226)	0.903	0.934
MNIST	256	0.105 (0.101–0.113)	0.224	0.115	0.921	0.934
MNIST	1,024	0.050 (0.050–0.050)	0.128	0.031	0.932	0.934
MNIST	2,048	**0.034** (0.031–0.035)	0.096	**0.024** (0.016–0.031)	**0.932**	0.934

How the width sweep scales:

- **On MNIST the network approaches its own linearisation as width$^{-0.51}$** (0.201 → 0.034 over 32× width). Its kernel drift falls as width$^{-0.60}$.
- **The infinite-width prediction is a worse description than the network's own kernel.** At width 2,048 the network is 9.6% from the infinite-width prediction but 3.4% from its own linearisation. The difference is the kernel's error at initialisation: the linearised and infinite-width predictions differ by 10.1% from each other.
- **Linearisation is a description for a window of time, not forever** (top middle of the figure). At every width the gap grows with training. On moons at width 1,024 the median goes from 0.004 at step 10 to 0.057 at step 1,000.
- **This explains the flat small-width end.** Moons' gap at the end of training barely changes from width 64 to 256 (0.123 → 0.114), even though at step 100 it halves (0.095 → 0.044).

The α sweep at width 512 (end of training; median of 3 seeds; log-log slopes fitted over α = 1–30):

	ReLU, moons	ReLU, MNIST	GELU, moons	GELU, MNIST
net vs linearised, $\alpha = 1$	0.074	0.081	0.018	0.040
net vs linearised, $\alpha = 30$	0.034	0.014	5.7×10^{-4}	1.3×10^{-3}
slope	**−0.23**	**−0.51**	**−1.02**	**−1.00**
kernel drift, α = 1 → 30	0.044 → 0.0071	0.079 → 0.0042	0.064 → 6.3×10^{-4}	0.090 → 7.9×10^{-4}
kernel drift slope	−0.51	−0.84	−1.36	−1.39
weight change, $\alpha = 1 \to 30$	$8.5 \times 10^{-3} \to 3.1 \times 10^{-4}$	$6.9 \times 10^{-3} \to 2.3 \times 10^{-4}$	$9.1 \times 10^{-3} \to 3.0 \times 10^{-4}$	$6.9 \times 10^{-3} \to 2.3 \times 10^{-4}$

The precision control (moons, width 512, seed 0, float32 against float64 weights):

	$\alpha = 30$, f32	$\alpha = 30$, f64	$\alpha = 100$, f32	$\alpha = 100$, f64
ReLU, net vs linearised	2.87×10^{-2}	3.23×10^{-2}	1.77×10^{-2}	1.73×10^{-2}
GELU, net vs linearised	5.7×10^{-4}	6.7×10^{-4}	2.4×10^{-3}	2.0×10^{-4}

Test accuracy and the convergence control:

- **MNIST.** At $\alpha = 0.03$ the networks reach **0.951** (ReLU) and **0.954** (GELU). With $\alpha \geq 1$ they reach 0.912–0.917 (ReLU) and 0.926–0.929 (GELU), and the infinite-width kernel at the same step reaches 0.934.
- **The kernel catches up when trained longer.** Trained to convergence, the infinite-width kernel model reaches **0.950** (0.950 already at 10× the steps, with train MSE 1.1×10^{-3}).
- **Moons.** Every arm sits at 0.964–0.974, and the converged kernel at 0.970.

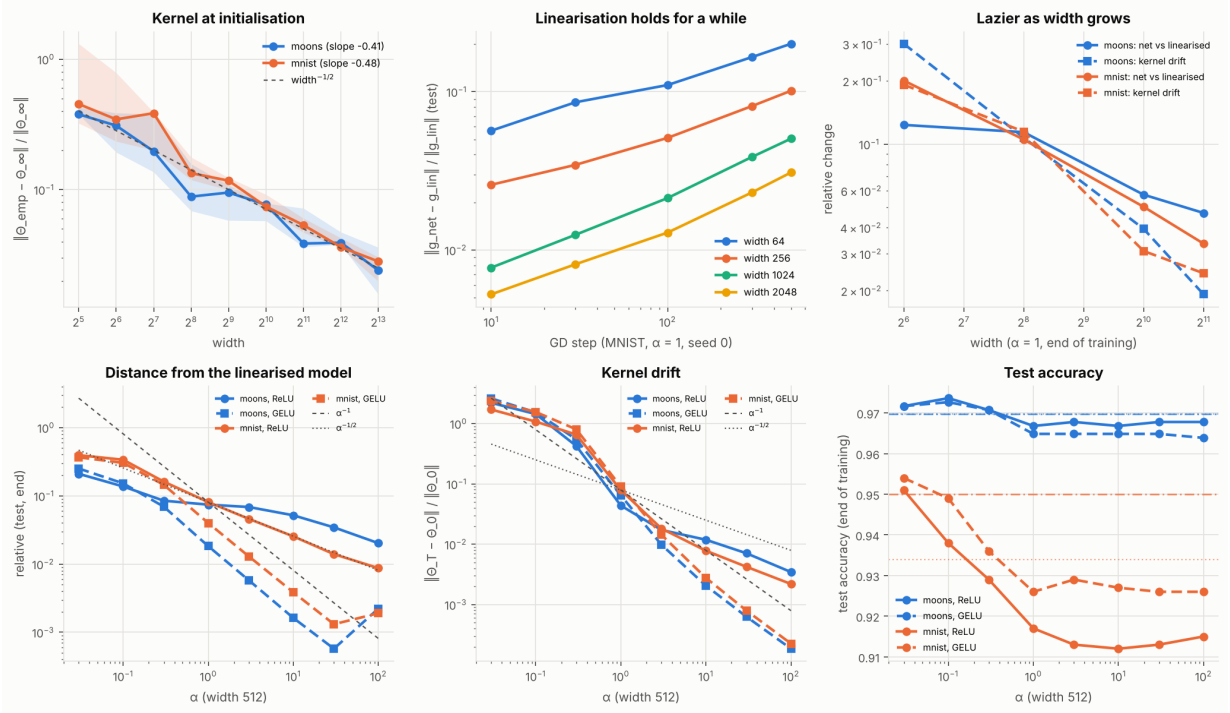


Figure 1: N7. Top: the empirical kernel at initialisation against width; the gap to the linearised model against training step for four widths; and the gap and kernel drift against width. Bottom: the gap and the drift against alpha for ReLU and GELU, with alpha^-1 and alpha^-1/2 guides; and test accuracy against alpha, where the dotted line is the infinite-width NTK at the same step and the dash-dot line the NTK trained to convergence.

Where the NTK picture holds, and where it doesn't:

- **"The kernel at initialisation converges as width$^{-1/2}$": holds.** The slope is -0.48 on MNIST and -0.41 on moons, the noisier of the two.
- **"Wide networks train as their kernel": holds, but for a window of time.** At width 2,048 the network is within 3.4% (MNIST) and 4.7% (moons) of its own linearisation. The gap grows with training at every width, so "lazy" means lazy for a number of steps that grows with width.
- **"The infinite-width kernel predicts the finite network": less well than its own kernel does.** At width 2,048 on MNIST the infinite-width prediction is 2.8× further away (9.6% against 3.4%).

- **"Lazy-training error falls as $1/\alpha$": holds for GELU, fails for ReLU.**
  - GELU gives slopes of -1.02 and -1.00.
  - ReLU gives -0.51 on MNIST and -0.23 to -0.35 on moons, with kernel drift near $\alpha^{-1/2}$.
  - This is not precision. Float64 weights give the same size of gap (1.73×10^{-2} against 1.77×10^{-2} at $\alpha = 100$). They are a different random draw, so the two are different networks, not the same network rounded differently.
- **"Feature learning beats the kernel": not shown here.** Small α moves the kernel by 170–260% and gives the best accuracy at a fixed step count. But the kernel model trained to convergence matches it (0.950 against 0.951 and 0.954).

What surprised me

ReLU is not lazy the way the theorem says.

- Chizat and Bach's $1/\alpha$ bound assumes a Lipschitz Jacobian. A ReLU network's Jacobian jumps whenever a unit's pre-activation crosses zero.
- The weights move by $\propto 1/\alpha$ (measured: the weight change falls exactly as $1/\alpha$). That flips a fraction $\propto 1/\alpha$ of the units for some input.
- Each flip changes the kernel by $\pm O(1/n)$ in no particular direction, so the changes add up incoherently. The total grows as $\sqrt{\#\text{flips}} \propto \alpha^{-1/2}$.
- The smooth control removes the effect. Swapping ReLU for GELU, with nothing else changed, restores $1/\alpha$ to two digits of slope on both tasks.
- So the measured ReLU slopes, -0.5 and shallower, are the kinks' signature. The theorem's premise was doing real work.

The first explanation was wrong twice.

- **The flat ReLU curve:** my first guess was float32. A float64 rerun gave the same size of gap (1.73 against 1.77×10^{-2} at $\alpha = 100$).
- **GELU's upturn at α = 100:** this one *is* float32. There, each step changes the weights by about 10^{-7} relative, float32's resolution. Float64 puts the point back on the line (2.0×10^{-4}, against 6.7×10^{-4} at $\alpha = 30$, a ratio of 3.3 for 3.3× in α).
- Same symptom, opposite causes, and only a rerun told them apart.

Feature learning looked like an accuracy win until the kernel got the same budget.

- At 500 steps, the small-α networks are 2–4 points ahead on MNIST.
- They have also driven train MSE to 0.001, against 0.135 for the lazy ones. Their kernel grows as the features move, so they train faster.
- Given 10× the steps, the infinite-width kernel reaches the same 0.950.
- On this task, at this scale, what feature learning bought was speed, not a better function. That is a statement about 1,000 MNIST digits and two hidden layers, not a general one.

The float32 arccos trap. On the kernel's diagonal $\cos\theta = 1$, where arccos has infinite slope. Float32 rounding of 10^{-7} in $\cos\theta$ becomes 3×10^{-4} in θ, which left the analytic NTK's diagonal 7×10^{-4} off. The comparison with `neural-tangents` caught it, and the kernels are now computed in float64.

Open threads

- **Measure the kink mechanism directly.** Count the activation flips per α and check that the kernel change is $\propto \sqrt{\text{flips}}$. Also check whether the ReLU exponent depends on width, as

an incoherent sum over n units suggests.
- **A fair accuracy comparison.** Train every α to the same training loss, not the same step count, with η tuned per arm. That is the test of whether features help at all here.
- **Wider α at small width, and the NTK eigenspectrum.** Which kernel eigendirections the trained kernel grows into is the usual way to see what "learned features" means.
- **Depth.** Two hidden layers only. The NTK's deep limit degenerates, and how fast finite networks follow it is the next question.

O7 — Mel front-end parity

Question

ml-ladder's part I left one thread open. With the prompt and the timestamp rules fixed, candle's whisper example still disagrees with transformers on 17 of 61 `tiny.en` transcripts (44 agree, temperature fallback off), and candle's production executor lands about as far away (41 of 61). The two share only the model weights and `pcm_to_mel`. Is the front end the cause? If so, which stage departs, and by how much?

The idea

Diff the front end stage by stage, then swap it under a fixed decoder.

- A feature-level diff says *where* two front ends differ. It cannot say whether the difference matters, because a recogniser may or may not be invariant to it.
- So the second half holds the decoder still. Every variant's features go through the *same* transformers `generate`, and only the features change.
- If transformers' decoder on candle's features reproduces candle's transcripts, the front end explains the disagreement. Whatever it does not reproduce belongs to the decoder.

candle runs as itself, not as my reading of it. candle's `whisper/audio.rs` is vendored byte for byte from the fork at the commit ml-ladder's runs used. It is compiled behind a small CLI, because the fork's `candle-transformers` does not build without CUDA. The `jnp` front end then gets candle's conventions as switches, and the switches are tested against the real binary.

The math

Whisper's features (openai-whisper, which transformers reproduces):

- reflect-pad the 30 s waveform by 200 samples at each end, so frame i is *centred* on sample $160i$;
- apply a periodic Hann window w of 400, compute $P_i(k) = |\mathrm{rfft}(w \cdot x_{160i-200\,:\,160i+200})(k)|^2$ for $k = 0..200$, and $m_i = FP_i$ with Slaney filters $F \in \mathbb{R}^{80\times 201}$;
- take $\ell = \log_{10} \max(m, 10^{-10})$, then $\widehat{\ell} = \max(\ell, \max \ell - 8)$, and the features are $(\widehat{\ell} + 4)/4$.

candle (copied from an early whisper.cpp) differs in two places:

1. **No centring.** Frame i is $x_{160i\,:\,160i+400}$. That is exactly the centred frame i of the audio *advanced by 200 samples* (12.5 ms), for every $i \geq 2$; frames 0 and 1 of the centred version reach into the reflect padding.
2. **Mirror folding.** It adds $|X(400-k)|^2$ into bin k for $k = 1..199$. For a real signal that is $2P(k)$, while DC and Nyquist stay single. Every Slaney filter is zero at both of those bins,

so $m \to 2m$ *exactly*. The log moves by $\log_{10} 2$, the floor $\max \ell - 8$ moves with it, and every feature shifts by $\log_{10}(2)/4 = 0.0753$. On the waveform, that is a gain of $\sqrt{2}$ (+3 dB).

So, up to frames 0 and 1, candle's front end *is* Whisper's, applied to audio that is 12.5 ms earlier and 3 dB louder. Both are perturbations a robust recogniser should not notice, so the question becomes empirical: is `tiny.en` robust here?

What we measure and why

- **Clips:** ml-ladder's 61, rebuilt.
  - the 50 FSDD recordings of speaker `yweweler` (digits 0–9, takes 0–4);
  - ten four-digit sequences from `default_rng(0)`, with 0.25 s gaps at 8 kHz;
  - all resampled 8 → 16 kHz with `resample_poly` and rounded through int16, as a wav file would be;
  - and JFK.

  The check that they are the same clips: transformers' `generate` on them reproduces ml-ladder's recorded transformers transcripts word for word on 61/61, and token for token on 60/61. The one token difference is a comma on `seq_5`.
- **Stage diffs** against transformers' extractor (its torch path, which is the one ml-ladder ran):
  - window and filters against float64;
  - power and mel as relative max error;
  - log and features as absolute error.

  candle exposes only its output, so its stages are those of the replica, once the replica is shown to equal it.
- **Decode:** transformers `tiny.en`, `generate(return_timestamps=True, max_new_tokens=200)`, greedy. This is ml-ladder's `whisper_hf_timestamps.py` protocol, the reference behind the 44/61. The scores are the following:
  - transcripts whose normalised words equal transformers' (ml-ladder's `words_of`);
  - the same against candle's example, using ml-ladder's recorded `prompt+pair-greedy` arm;
  - digit exact-match, and sequence and JFK WER.
- **Controls**, through transformers' *own* extractor:
  - candle's conventions as waveform transforms (advance 200 samples; gain $\sqrt{2}$);
  - the same kind of perturbation at other sizes (a one-frame delay of 160 samples, half-frame shifts of 80, a gain of $1/\sqrt{2}$).

  These calibrate how much any 5–12 ms shift or 3 dB change moves these transcripts.
- **No seeds.** Every step here is deterministic. The perturbation controls play the role a seed spread plays elsewhere.

What I built

- `jaxexp.mel`:
  - `mel_filters` (Slaney scale and normalisation in NumPy);
  - `stages` / `log_mel` in `jnp`, with `center` and `double_bins` switches.
- `tools/candle_mel`: candle's `audio.rs`, vendored verbatim (SHA-256 checked at run time), plus a CLI that reads PCM and filters and writes the features. `vendor/README.md` records its source, commit, hash and licence.
- `experiments/o7_mel/run.py`, with four parts:
  - `stages`: five front ends on 61 clips;
  - `decode`: nine feature variants through one decoder;

  - `controls`: seven waveform perturbations;
  - `plot`.
- `tests/test_07_mel.py`, five tests:
  - candle's filter file matches `mel_filters`;
  - `log_mel` matches a float64 NumPy Whisper to 10^{-4};
  - `double_bins` is a uniform $\log_{10}(2)/4$;
  - uncentred frames start at $160i$;
  - with both switches on, `jnp` reproduces the real candle binary to 5×10^{-4}.

Results

Stage by stage: everything agrees except candle, and candle is two switches.

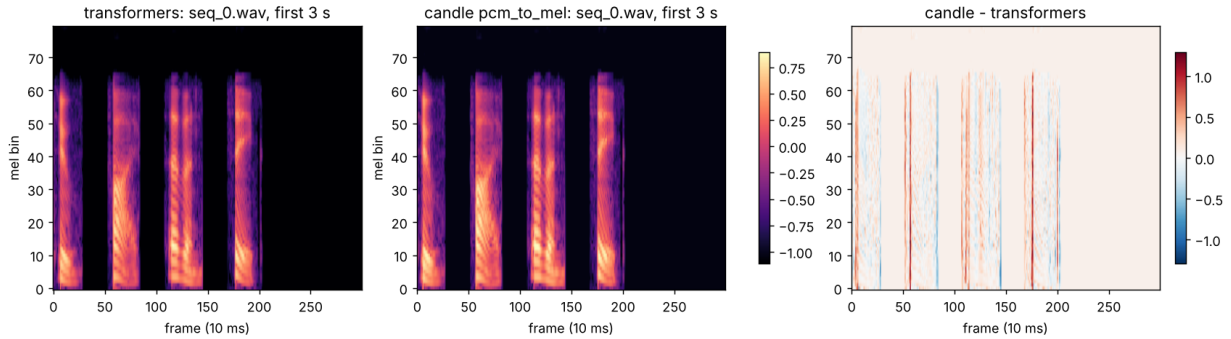


Figure 2: candle's pcm_to_mel against transformers on one sequence: onset and offset edges (the time shift) on a uniform offset (the gain)

stage, against transformers	torchaudio	`jnp`	transformers' NumPy path	candle
window (vs float64)	2.5×10^{-7}	2.8×10^{-8}	—	2.5×10^{-7} (same formula in f32)
filters (vs float64; max weight 0.026)	5.7×10^{-8}	9.2×10^{-10}	—	1.3×10^{-9} (`melfilters.bytes`)
power, relative	0 (same `torch.stft`)	4.0×10^{-7}	—	frames shifted by 200 samples, bins ×2
mel, relative	2.2×10^{-6}	3.7×10^{-7}	—	×2 on every value
features, max abs	7.6×10^{-6}	5.0×10^{-5}	4.4×10^{-5}	**1.15** (0.79–1.51)

Worst clip of 61, except candle's features: median over clips, with min–max.

- **The replica is candle.** `jnp` with both switches and candle's filter file matches the binary to 4.1×10^{-5} (median), and 1.5×10^{-4} on the worst clip. That residual is all that is left for candle's recursive f32 FFT and its thread-split loop. Both are numerically harmless.
- **The filter file is not a suspect.** It matches transformers' float64 filters to 1.3×10^{-9}, and swapping it into `jnp` changes no feature by more than float32 round-off (identical 5.0×10^{-5} worst case).

- **Size of each switch,** as mean $|\Delta|$ over speech frames, median and range over clips:
  - centring: 0.107 (0.075–0.165), with up to 1.08 at onsets and offsets;
  - bin doubling: 0.0753 on every clip and every feature, the predicted $\log_{10}(2)/4$;
  - both together: 0.127 (0.108–0.169).
- **The equivalence holds.** transformers' own extractor on audio advanced by 200 samples matches the uncentred features to 2.9×10^{-4} from frame 2 on.

Under a fixed decoder, candle's features reproduce candle's transcripts.

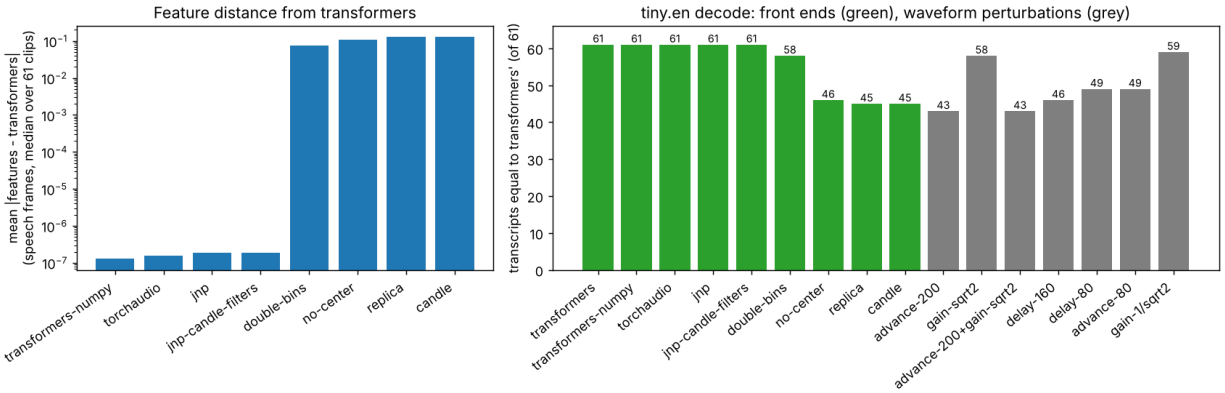


Figure 3: Feature distance and transcript agreement for each front end and each waveform perturbation

features into transformers' `tiny.en`	equal to transformers' (of 61)	equal to candle's example (of 61)	digits exact	sequence WER	JFK WER
transformers (torch path, numpy path), torchaudio, `jnp`, `jnp` + candle's filters	61	44	0.44	0.150	0
bins doubled only	58	44	0.48	2.625	0
uncentred only	46	56	0.40	2.625	0
candle's `pcm_to_mel` (= replica)	**45**	**59**	**0.44**	2.650	0
candle's own example (ml-ladder, recorded)	*44*	—	*0.44*	*0.200*	*0*

- **The front end explains 15 of the 17 disagreements.** candle's example and transformers differ on 17 clips. Fed candle's features, transformers' decoder moves to candle's answer on 15 of them, and moves away on none. It even lands on candle's digit scores exactly: 0.44 exact and 0.84 WER, against 0.44 and 0.82 on transformers' features.
- **Two clips are the decoder's**, both sequences:

  - on `seq_0`, candle's decoder hears one more digit ("2 5 7 6" against "5, 7, 6.");
  - on `seq_1`, transformers' decoder on candle's features falls into a repetition loop ("1, 2, 3, 3, 3, ...", to the 200-token cap), where candle's stops at "1, take.". candle's decoder is greedy too (ml-ladder ran it with the fallback off), and its token cap of 224 is above the 200 used here, so the cap is not why it escapes.

  That one clip (WER 25) is the whole of the 2.65 sequence WER in the table. Without it, candle's features score 0.167 on the other nine sequences, against 0.139 for transformers' features and 0.139 for candle's example.
- **Centring is most of it.** Alone it moves 15 transcripts, against 3 for the bin doubling.

And the decoder is that fragile anyway.

waveform perturbation, transformers' own extractor	equal to transformers' (of 61)	digits exact	JFK WER
gain $1/\sqrt{2}$ (-3 dB)	59	0.46	0
gain $\sqrt{2}$ (+3 dB; candle's doubling)	58	0.48	0
delay 80 samples (5 ms)	49	0.48	0
advance 80 samples (5 ms)	49	0.48	0
delay 160 samples (one frame, 10 ms)	46	0.48	0
advance 200 samples (12.5 ms; candle's framing)	43	0.42	0
advance 200 and gain $\sqrt{2}$	43	0.42	0

- **A pure one-frame delay**, which moves every feature column by one position and changes nothing else, flips 15 of 61 transcripts. candle's front end flips 16.
- **Accuracy does not move.** Digit exact-match stays between 0.40 and 0.48 under every perturbation, around transformers' 0.44, and JFK is perfect in every arm.
- **Even the first two frames matter.** "Advance 200" and "uncentred" differ only in frames 0 and 1, and they disagree on 4 transcripts (43 against 46 equal to transformers'; the first matches the candle-feature transcripts on 56).

What surprised me

The first suspect was right, and it still is not a bug that costs accuracy. candle's front end is the whole disagreement: 15 of 17 transcripts follow it when nothing else changes. But what it does to the signal is a 12.5 ms time shift and a 3 dB gain. On these clips, `tiny.en` responds to a 10 ms delay of perfect audio about as strongly. Single digits from one speaker, upsampled from 8 kHz, sit near `tiny.en`'s decision boundaries, so its transcripts there are chaotic at the 10 ms scale while its accuracy is not. The 17/61 "disagreement" is real and reproducible, but it does not show candle transcribing worse.

Doubling every bin turns out to be a constant. I expected the mirror fold to distort the spectrum. Because Slaney filters vanish at DC and at Nyquist, the two bins the fold leaves single, it doubles every mel energy exactly. The log turns that into a single additive constant, and the clamp's floor moves with the max, so it stays constant. Its 0.0753 shift is the same on every feature of every clip, to four digits.

The equivalence needed two frames, not one. I first excluded only frame 0 from the "advance by 200 samples" check and found a 0.43 gap. Frame 1 of the centred version starts 40 samples before the shifted audio, so it also sees reflect padding. From frame 2 on, the gap is 2.9×10^{-4}.

Open threads

- **The executor's 41/61 is not re-measured here.** It resamples the raw 8 kHz files itself with a linear resampler, so its input differs from these clips before the front end does. Rebuilding its resampler is the step that would close that number.
- **The two decoder-side clips.** `seq_0` and `seq_1` differ between candle's and transformers' decoders on identical features. One is a repetition loop that candle avoids at a larger token cap, so its suppression and timestamp rules are the first place to look. f32 numerics come second.
- **A patch for candle**, ready to send with the fork's example fix (davidmiheev/candle#13):
  - centre the frames, by reflect-padding 200 samples at the start;
  - drop the fold.

  The `jnp` switches show what the patched output must match (to 5×10^{-5}). Not checked: whether upstream candle or whisper.cpp has changed `audio.rs` since the fork's commit.
- **A robustness metric for Whisper ports.** Agreement with a reference implementation is a poor parity test on clips where the reference itself flips under a 10 ms shift. Agreement on a clip set where one-frame delays change nothing (JFK-like speech) would separate real port bugs from this fragility.